

Solver gap engine

Ranking open math problems by closeness to solved ones, with a Lean proof and empirical check path.

Result

Measure	Value
Problems mapped	3,599 from formal-conjectures at commit 7d450ef
Solved, open	2,196, 1,403
Solved with a Lean proof	783
Open problems ranked	1,364 with a docstring over 30 characters
Cross-file similarity 0.9 or above	5 [empirical: match.py stats.json, 2026-09-30, python3 match.py]
Cross-file similarity 0.8 or above	31
Median nearest cross-file similarity	0.586
Closest match in the same file, excluded	455

Method

Problem statements come from the `@[category research open|solved]` tags. Docstrings are embedded with all-MiniLM-L6-v2 and each open problem is matched to its nearest solved problems in other files, since same-file matches are variants of one problem [empirical: `_intake/solver-gap-engine/match.py`, 2026-09-30, python3 match.py].

The Lean test file proves the odd-sum identity, Fermat's little theorem and Wilson's theorem, then samples all three plus Goldbach with zero failures, and exits 1 on any failure [empirical: `harness/Empirical.lean`, 2026-09-30, lake env lean --run].

Reproduction steps and pinned versions: `_intake/solver-gap-engine/README.md`.

Limits

Similarity measures wording. A high score marks a candidate for a prover and says nothing about difficulty.

Next

1. Run a neural prover on the top 50, bead bkt-wfz9, blocked on a GPU host.
2. Replication funding for empirical checks, spec in `_intake/solver-gap-engine/REPLICATION.md`.
3. Zero-knowledge receipts for gold-layer claims under the medallion layers, bead bkt-etmx.